

ActInf GuestStream #025.1 ~ "Autistic-Like Traits, Positive Schizotypy, Predictive Mind"

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hello everyone it is august 3rd 2022 and it's octave guest stream number 25.1 we're here with brett anderson and we'll be hearing about autistic-like traits and positive schizotypy as diametric specializations of the predictive mind there will be first a presentation followed by a discussion segment so if you're watching live feel free to add any comments or questions and brett thank you again for joining and on with the presentation all right thank you daniel uh for having me so yeah this is a paper that was published this earlier this month and uh so before i get into the paper i should probably introduce myself just a little bit so i am an evolutionary psychology phd student at the university of new mexico my research mainly has to do with something called the diametric model of autism and psychosis which i'll talk more about that here in just a minute so i am not trained in active inference or predictive processing i came to this because i was looking for a theory that could explain variation along this autism schizotypy continuum and i'll i'll say what that is here in just a minute and i came across uh santa croix or let's see sanders van der kroys and colleagues 2014 their paper about autism published in psychological review and it was through that paper that i sort of came to write this paper because uh i realized at some point that that was the theory i was looking for and that uh the opposite configuration of what they put forward so they said that autism results from giving an inflexibly high weight to sensory input and what i realized through reading that paper and reading some other stuff around the topic was that giving an inflexibly low weight to sensory input does a really good job of explaining the other side of that continuum and so that's what this paper is about so i'm putting forward an explanation for a for a continuum of individual differences so uh just an idea of what i'm going to talk about here so i'm going to talk about the diametric model of autism and psychosis what it is what some of the evidence for it is how it manifests in non-clinical populations which is what my research mainly focuses on and we refer to that as the autism schizotypy continuum uh and then the hypothesis that i put forward in this paper which has to do with explaining variation along this continuum uh

due to trade-offs in precision weighting due to precision weighting and and then i'll talk about some implications for predictive processing and active reference so the diametric model of autism and psychosis was first put forward by crespì and babcock uh bernie crest bernie crespì and christopher babcock in a 2008 bbs paper it is very simply uh very simply put just the idea that autism spectrum conditions and psychosis spectrum conditions are cognitive and genetic opposites before i get into some of the evidence for it i want to at least address what i think is the most common objection i hear it all the time which is that how can the diametric model be true given that some people who are diagnosed with an autism spectrum condition experience psychosis and vice versa right so there's there's comorbidity between these disorders how can this diametric model be the case be true and i have a whole video about that on my youtube channel i might put the link in the comments to this to this youtube video uh but basically the response to this objection is just to point out that dsm categories are extremely heterogeneous so within each dsm category you're going to find multiple etiologies that may be totally unrelated to each other in some cases so for example you could have somebody diagnosed with an autism spectrum condition uh is is diagnosed with that mainly because for example they have like a language condition they have maybe a language delay and that causes a communicative a communication problem and then they get the autism diagnosis and it's not supposed to happen that way but it definitely does happen that way and so that would be different from somebody who has a more classical presentation with restricted and repetitive behaviors and so on and so that person their their etiology may have nothing whatsoever to do with the diametric model so basically given the heterogeneity of these conditions we don't expect to have any precise theory that's going to explain uh every case of the condition and that includes the diametric model so what is some of the evidence for the i think the best evidence for it is just to point out all of the diametrically related uh characteristics on either side of this what's what's meant to be a continuum so with autism spectrum conditions there's evidence that paternally expressed genes promote autism so we get two copies of every gene we get one from our mom and one from our dad there are a subset of genes that are called imprinted genes and with those genes you only express one of those copies so you might express the copy from you only from your dad that's a paternally expressed gene paternally expressed genes when they're hyper expressed tend to promote autism autism spectrum conditions we see overgrowth in autism especially in infancy hypomentalizing which is reduced attribution of mental states to other people reduced imagination hyper systemizing relative right hemisphere dysfunction increased cancer rates which is related to the genomic

imprinting issue and we also see a sex difference where men tend to be higher in autistic like traits and they tend to be diagnosed with autism more often and on the psychosis side of things we see a lot of opposite characteristics so we see evidence that maternally expressed genes promote psychosis spectrum conditions we see undergrowth especially in infancy hypermentalizing which is increased attribution of mental states to other people often that that manifest as paranoia increased imagination apophenia which is predisposition to false positives relative left hemisphere dysfunction reduced cancer rates at least with schizophrenia and we also see a sex difference with positive schizotypes so women tend to be higher in positive schizotype we're talking about psychosis so men tend to be so the type of psychosis that men tend to present with tends to be more often degenerative schizophrenia uh whereas women tend to uh present more with acute psychosis which seems to have more of a relation to the diametric model so um now what this paper is focused on is how the diametric model manifests in non-clinical population so we we talk about autistic like traits on one end of this continuum and positive schizotype on the other and so the hypothesis put forward in this paper is not primarily about dysfunction it's not primarily about explaining mental condition mental disorders or mental conditions it's about explaining a continuum of individual differences that we all fall somewhere along and so this is the this is referred to as the autism schizotheric continuum on the autistic side on the uh side associated with autistic like traits we see hypomentalizing reduced imagination increased systemizing they tend to be relatively poor at noisy pattern detection although they're very good at pattern detection when there's not a lot of noise in the system we see this narrow focused attention style and people high in autistic like traits are over represented among engineers and other technical professions on the positive schizotype side we see hypermentalizing increased imagination reduced systemizing apophenia and this more broad distractible attention style and they tend to be over represented among artists and other creative professions and so the hypothesis that i put forward to try to explain variation along this continuum i refer to it as the precision weighting of sensory input hypothesis uh as i'm sure everybody listening to this knows precision weighting underlies the allocation of attention you can adjust the precision that you're giving to sensory input in order to account for noise in the input and van der kroys and colleagues in their 2014 paper published in psychological review uh this was not the only so there was a few papers published around this time that were sort of talking about this idea theirs was the first to really put it forward and like in full detail so they argued that autism spectrum disorders are caused by giving an inflexibly high weight to sensory prediction errors

and my paper simply extends this hypothesis i'm arguing that there is a continuum of individual differences characterized by people who give a relatively high weight to sensory input on one side that's associated with autism and autistic electrodes and people who give a relatively low weight to sensory input on the other side which is associated with positive schizotyping and these differences are going to be associated with uh over with with sort of leaning towards overfitting sensory input which is what happens in autism and leaning towards underfitting sensory input and that under fitting can explain some of the features of positive schizotyping and so just to give a visual demonstration of this there's a very simplified visual demonstration but the dots here are sensory input right and the line is the model so if you're overfitting if you're leaning towards giving a high weight to sensory input you're going to end up fitting your model too precisely to the data and that means that you're going to incorporate so a lot of that variation in the model is noise you're going to end up incorporating noise into your model the opposite problem that you can have is instead of giving too much weight to the sensory input you're giving too much weight to your hypothesis about the input so you're kind of imposing your hypothesis onto the model you're imposing your prediction onto the input so in this example on the right hand side of the screen the top down prediction is that the data is linear and if you give too much weight to that prediction you can end up imposing a linear model onto the data something like this i will argue is going on in positive schizotype and then there is some optimal balance that you're going to find that is going to give you the best chance of finding the pattern underlying the data importantly that balance is is never the same or it's not it's not stable so it depends on how much noise is in the data depends on how how much you want to generalize from the data and so there is no single optimal solution right there are always trade-offs associated with this these are the differences that i am i'm going to argue that i can explain with this hypothesis so differences along the autism schizotype continuum these include bottom-up versus top-down influences on perception and cognition hypomentalizing versus hypermentalizing reduced imagination versus increased imagination and everything everything on the left here is associated with the autistic like traits and then narrow detailed attention versus broad distractible attention apophenia which is associated with positive schizotype systemizing which is associated with autism and autistic like traits and then differences in the propensity for exploration so we're going to go through all of these and show how this hypothesis may be able to account for them with bottom-up versus top-down influences on perception uh crespi and dinsdale in a 2019 paper reviewed evidence for diametric susceptibility to the rubber hand illusion so most

people probably know what the rubber hand illusion is got a little visual thing on the right hand side that shows what it is so with the rubber hand illusion you have your real hand hidden by some screen or something and a rubber hand is placed in front of you where you think the real hand is going to go the experimenter will take a feather or something and they will stroke your real hand at the same time as they stroke the rubber hand and that will produce the illusion that the rubber hand belongs to you and oftentimes at the end of the experiment they'll take a hammer and slam it down on the rubber hand and you'll jump and all that so you really you really do sort of think that the hand is yours uh what crispy and dinsdale reviewed evidence for is that oh um before i get to that so jacob howie i think that's how you say that say his name uh jacob howie in his book suggested that the illusion works because of the top-down prior belief that systematically related properties in this case seeing a hand being stroked and feeling your own hand being stroked at the same time are very likely to be causally related and so what that means is that susceptibility to the illusion should be facilitated by by giving more weight to your top down prior so a relative overweighting of top-down priors should facilitate susceptibility to the illusion and decreased susceptibility should be facilitated by giving more weight to bottom-up sensor input and so in that case we would expect for autism and autistic like traits to be associated with reduced susceptibility and positive schizotypy and psychosis to be associated with more susceptibility and that is what crespi and dinsdale find so they find that autism is associated with reduced susceptibility and vice versa for positive schizotypy and uh predisposition to psychosis moving on to mentalizing within the predictive processing framework uh mentalizing our theory of mind is thought to result from the use of relatively abstract or high-level predictions uh priors to predict other people's behavior and so again sort of the same principle as the last time if that's the case we would expect an overweighting of bottom-up input relative to priors should result in hypomentalizing or other people and an overweighting of priors should result in the opposite of hypermentalizing or giving an increased attribution of mental states to other people and of course that is what we find associated with the autism schizotypy continuum uh this was one of the things that van der kroes and colleagues talked about in their 2014 paper but the opposite positive schizotypal does result in the opposite finding so next up is imagination uh andy clark in a 2013 paper he argued that we we go into our imagination when we should when we sufficiently turn down the weight that we're giving to proprioceptive sensory input and so when we do that we are now dealing with our top down prediction sort of offline right unmoored from the sensory input what this should mean is that if you're giving an inflexibly

high weight to sensory input that should make it more difficult to turn down that weight enough to go into your imagination if you're giving an inflexibly low weight to sensory input this should mean that you will find it easier to go into your imagination and that is we do find this difference along the autism schizotypal continuum so people high in autistic like traits uh they have imaginative deficits often and people high in positive schizotype they do better on tasks that at least purportedly measure imagination like divergent thinking tasks there's evidence that they spend more time daydreaming and mind wandering and of course they're also over represented among artists and creative professions uh moving on to attention abu aakl and colleagues in a couple papers in 2017 and 2018 demonstrated diametric attention styles for people high and positive schizotyping and people high in autistic like traits what they showed especially in the in the 2018 paper uh people high in autistic like traits do better on a task where they had to ignore salient distractors right so the task was essentially they had to focus on something on the screen and they had to ignore salient distractors that would pop up in the periphery and that facilitated them doing better on the task on the other hand people high in positive schizotype did better on a task that was facilitated by the opposite by being distracted by the by the things in the periphery uh so we see this sort of focused attention style with autism and we see this uh more diffuse distractible attention style with positive schizotype there is other evidence for this and i review that in the paper if you want to look at that the question is how can the psi hypothesis account for this i think this one even though it should you might think it's the most straightforward one because precision weighting is about attention and this is attention it's actually i think the least straightforward one and the reason why is because it does seem somewhat intuitive that if you're being bombarded sort of by highly precise sensory prediction errors which is what we argue happens in autism uh you might think that would make you more distractible and vice versa for positive schizotypes since they're only paying attention to relatively large prediction errors that might make them less distractible that is that isn't the case but it'll take a little bit of unpacking to explain why and i'm going to use an analogy here we're going to use what i'm calling the fire department analogy so you can imagine that you have two fire departments we'll just call them fire department a and fire department b and there are three kinds of fires that you could have so you could have small fires medium fires and large fires and these departments receive calls about the fires but they are not told how big the fires are right so they get a call they're told that there's a fire at this location but they're not told whether it's a small medium or large fire they're just told that there's a fire department a has a policy where they they receive calls about

all kinds of fires right so they receive calls about small medium and large fires and department b has a policy where they only receive calls about large fires right so they don't take calls about small and medium fires i'm arguing that this is basically equivalent you know within this analogy to giving a high weight to sensory input for department a which they take calls about all fires and as van der kroy's and colleagues pointed out in their 2014 paper when you're giving a high weight to sensory input you're treating all of these sensory prediction errors as if they're all important the problem with that of course is that not all sensory prediction errors are actually important department b who only receives calls about large fires this is equivalent to giving a low weight to sensory input so which strategy works better that depends on the context so if there are more fires than you can put out at a given time right if the whole city catches fire at once department b works better because in that case you actually don't want to pay attention to the small fires where you only want to pay attention to the large fires you want to ignore the details uh if there's not that many fires at a time you don't want to ignore small fires right you want to pay attention to the small fires so department a works better in that context now imagine that these two departments are in the midst of putting out a particularly large and important fire they receive a call about a new fire and the question here is what should they do in relation to this call department a should maintain focus on the current fire that's because when they receive a call they don't know whether it's small medium or large it's more likely to be small or medium so they should ignore the call maintain focus on their current task department b should be more willing to allocate resources to the new fire right they know that it's a large fire because they only receive calls about large fires so they should be more willing to redirect their attentional resources okay this is my argument for why we see this uh attentional difference along the autism schizotypal continuum uh people high and autistic like traits have learned that the peripheral prediction errors are very likely to be unimportant and that they should focus on the task at hand that learning probably happens unconsciously uh people behind positive schizotype have learned the opposite right if they're if they receive a prediction error on the periphery they've learned to assume that it's important right because they only pay attention to important prediction areas so uh so anyways this this analogy is what i'm using to explain this attentional difference along the autism schizotype uh so moving on to systemizing so people high in autistic like traits and people with with an autism spectrum condition are often very skilled at learning about uh rules-based systems so this would be something like a computer programming language most of mathematics falls under this category uh they tend to be very good at these things if you'll remember you know

overfitting results from incorporating noise into your predictive model the thing about a rules-based system is that there is no noise in that system if it's truly rules-based then when you put in a certain input you get the same output every time it can be an extremely complex these can be extremely complex rules uh like there are with computer programming languages but they're extremely precise and so that highly precise cognitive perceptual style that comes with giving a high weight to sensory input is going to facilitate the systemizing ability with autism and autistic like traits so for a visual demonstration of this you can have this complex pattern and there's no noise in the pattern right it might be highly complex but there's no noise if you're giving a high weight to sensory input you can fit your model as precisely as you want to this pattern and you're not going to run into the problem of overfitting because there is no noise to be incorporated in the model people high in positive schizotypy on the other hand may attempt to impose a simplifying assumption onto the model right they may attempt to impose simplifying assumptions onto this complex but non-noisy input and that will lead them to not be as good at systemizing and that is what we find looking at noisy patterns now so people with autism spectrum conditions are relatively poor at picking up on noisy patterns i was not able to get to this in the paper because i didn't have room but it is something that i would have included if i had room uh we reviewed the evidence for this in an upcoming paper with myself john vervicky and mark miller so if you're giving a high weight to it's going to make it more difficult to pick up on patterns that were that are embedded in noise and there is no evidence that people high in positive schizotype have issues with noisy patterns so apophenia has a different thing we'll talk about the next but but there is evidence for this with autistic like traits and asd we can understand why that's the case for the same reason so if you have this noisy data and you're trying to fit your model very precisely to that noisy data you are not going to pick up on the underlying pattern right you're going to miss out on the underlying pattern you're going to over fit your model to this data so people high in positive schizotypy really ought to be better at picking up on these kinds of noisy patterns because they're not going to miss out on the big picture by paying too much attention to the details is essentially what would happen there so apophenia is the next one so apophenia is when you perceive patterns and pure noise right and this is thought to be a core feature of positive uh so for example people high in positive schizotypal are more likely to hear messages if they're if you listen to audio that's pure noise they're more likely to say that they heard a voice in the noise or something like that and giving a low weight to sensory input means that they will often impose their top down assumptions onto pure noise there's no

evidence that people high in autistic like traits have issues with apophenia and so again this is just a visual demonstration of this um with autistic like traits when they're presented with pure noise again they're not going to see any pattern in that right they're just going to see randomness people high in positive may again they may attempt to impose a pattern onto this data because they're giving more weight to the top down to the top down assumption and less weight to the bottom up input so i think this is the last one i might finish a little early exploitation exploration so we do see uh evidence okay so yeah so deciding whether to exploit current opportunities or search for new ones direct exploration is a fundamental trade-off that faces all organisms and there is evidence that people high and positive schizotypal are more exploratory than people high in autistic like traits so for example they are higher in the personality traits of openness and extraversion uh colin de young and some of some of his colleagues have argued that those two traits form a higher order trait called plasticity and the young has argued that that plasticity represents the general tendency towards exploration so that's uh some evidence uh for differences in exploratory tendency we also see differences in sensation seeking associated with this so sensation seeking has also been argued to be an indicator of exploratory tendency kirsten and colleagues and 2019 paper said this they they said that a large but resolvable error signal informs the agent that the environment offers the opportunity or offers the opportunity to resolve uncertainty that opportunity to resolve uncertainty makes exploration an attractive option the issue is so if you're if you're giving an inflexibly high weight to sensory input what that means is that you're not seeing large differences between the prediction errors that are that you're receiving means that people high in positive schizotyping only tend to pay attention to large prediction errors and these large prediction errors are more likely to incentivize exploration they represent a greater potential information game information gain and so people high in high and autistic like traits we see that autism is associated with restricted and repetitive behaviors and relatively narrow interests uh and that me and that's a reflection of this reduced exploration yeah so like i said before giving a high weight to sensory input means that most prediction errors are going to be treated as relatively equal and in importance and as van der kroes and colleagues pointed out in the 2014 paper if each prediction error is as important as the next it is more difficult to estimate where predictive process can be made and therefore exploration is not as enticing this comes with a benefit however because the the decreased exploration that's associated with autistic literates is also why they tend to be more competent specialists they tend to specialize on one highly particular domain and sort of drill deep into one thing instead of spreading out their

options over many things they are less likely to be distracted by emergent opportunities what are some of the implications of this hypothesis uh so like i said van der kroys and colleagues their 2014 theory about autism should not be seen as being just a theory about autism but rather as a basis for a theory about individual differences in general because in addition to people who give an inflexibly high weight to sensory input i've argued here that there are people who also given inflexibly low weight to sensory prediction errors and those are people high in positive schizotyping and this uh just as a reminder this should not be thought of in terms of dysfunction it is a continuum of individual differences that we all fall somewhere along right so this is just a a continuum that that will occur in non-clinical populations as well and this paper was an initial attempt to sketch the evidence and implications of this idea there is still room for an empirical research program to test and extend this this idea and that's what i'm going to do with the rest of my phd most likely and theoretical work on its implications so i am an evolutionary psychologist so i should talk a little bit about some evolutionary implications the autism schizotypal continuum is a highly heritable source of individual differences we know that autism and schizophrenia for example are about have a heritability of about 0.8 so they're very heritable this means that they're going to be highly responsive to selection pressures and if this hypothesis that put forward here is correct my contention is that this continuum is very likely equivalent or at least analogous to specialist generalist continuums elsewhere in biology and that is because the the precise perceptual style and autism is going to facilitate specializing for highly particular domains we do see this with autism they tend to have relatively narrow interests but that allows them to to become very specialized in some some particular domain the imprecise perceptual style with positive schizotype is going to facilitate picking up on these more noisy patterns and noisy patterns tend to be more generalizable they tend to generalize across context a little bit better there are models from theoretical biology that have been used to understand how specialist generalist continuums like this can evolve and basically the you know if i was going to extract a general principle from these models uh what they what they tend to argue is that if you have periods of relative stability punctuated occasionally by periods of upheaval so if you have fluctuating selection pressures this can maintain a heritable continuum of individuals individual differences like this one and there is evidence that this was the environment that characterized human beings before the advent of agriculture so before the advent of agriculture there was a lot more fluctuations in the climate things were changing a lot more often so we could see the evolution of a continuum of individual differences or at least the maintenance of a continuum

like this one so that is the end and thank you to daniel thank you to the active inference institute for letting me share my work and that is it thanks a lot you can unshare and we'll talk a little bit more about this in our store all right well to those who are watching along live please feel free to add any questions in and if they're relevant we'll go with them well a lot there i wrote down a bunch of questions to enter the maze at just one point here we have two axes about the extent of the two traits that you're focusing on and the diametric model in my understanding summarizes a manifold and says that there's correlation across these two dimensions so i wondered what is that state space are these like the big two or what exactly else might be in this broader space of yeah it's a really good question and uh the thing is you know if we're looking at something like the big five right so this is how we normally think about individual differences we think about them in terms of the big five as a psychologist the thing about the big five is that it my argument would be that it doesn't carve nature at its joints right it's this very top-down way of understanding and it doesn't it's it's completely neutral about mechanism with this with the autism schizotype we're coming at things from the opposite direction right so it's a bottom-up uh way of thinking about individual differences and at least if this hypothesis is correct we have a mechanism right if this does explain the things i think it explains in it we have a mechanism to explain it now clearly and you know if it is just precision weighting given the importance of precision weighting within the predictive processing framework we would expect this to be a pretty important uh source of individual differences however there are clearly many other sources of individual differences that are going to interact with this in uh unpredictable ways so for example you know you're going to have you're going to have things that are that are mediated for example by blood testosterone levels right there are clearly individual differences mediated by testosterone levels i don't think this has much to do with that so how does that interact with this how do you know uh differences in something like agreeableness which i don't think has much to do with this you know how do these things interact it's it's a big question um my advisor and i are working on a scale to measure this stuff so that we can start unpacking some of those interactions and trying to figure some of that out but i would say that we're just at the very beginning of that of that process so you know sort of conclude on your quest on your question we don't really know i think that this is a large source of individual differences but in terms of how much of the variation and something like the big five trait of openness for example clearly this has something to do with variation and openness how much does it account for we also know that intelligence accounts for some of that variation how much does intelligence account for

compared to this these are all open questions that that we're still we still haven't answered yet awesome thank you sure when i was seeing the side-by-side traits i was really struck by the mention of cancer i am not an expert however these conditions in their clinical forms and potentially in their subclinical variances are associated with other organ systems like digestive system and skin and immunity and so on so i was just curious about what was the cancer empirical finding and how did this relate to multi-tissue multi-system effects and maybe even embodied cognition yeah yeah so it is the case that autism conditions are associated with an increased cancer rate uh with psychosis the evidence is a little bit more mixed so it is uh at least at the time that that cresspi and babcock wrote their paper there was good evidence that schizophrenia is associated with reduced cancer rates i've looked at some of the evidence more recently with schizophrenia i think that that that finding is still solid with bipolar disorder which is another psychosis spectrum condition uh the findings are definitely more mixed either way the the fact that schizophrenia is associated with reduced cancer is a very strange finding uh and the reason you know main reason why is because schizophrenia is pretty much associated with everything negative you can think of i mean the fact that it's it's associated with reduced cancer rates is strange just in light of that fact athena activists wrote a book about cancer in 2018 i wish i could remember the name of that book but in that book she talked about so she didn't talk about it in terms of autism and psychosis but she talked about it in terms of imprinted genes so i i said that autism is associated with paternally expressed genes and psychosis is associated with maternally expressed genes with imprinted genes what happens is that the genes that are that come from your dad they promote your dad's genetic interests now the thing about human beings is that we do not have complete paternity certainty meaning that it's not always you don't always know if you're a man you don't always know that the kids are yours right uh as a mother you know that all the kids are yours and you don't always know as a man they're gonna have kids with the same with the same woman so the paternally expressed genes have an interest in promoting growth they have an interest in taking as much of the mother's resources as they can and that is what they do and because they promote growth if they're hyper expressed they will also end up promoting cancer because cancer is something that uh that results from too much of you know the hormones that promote growth it can result from that on the other side with psychosis because it's associated with maternally expressed genes so as i said it's associated with undergrowth and infancy it's also associated with reduced suckling so kids who grow grow up to have schizophrenia they don't suckle as much they don't take as much of their mother's milk they also um there's

evidence that they also just tend to be more calm as children and this is the the idea of the theory here is that this is because they are promoting their mother's interests right they have more they're they're expressing their mother's genes more and that promotes their the things that are interest that are in the interest of the mother's genetic interests so that would be reduced growth reduce suckling being an easier baby they take care of and so on uh so that's the that's the the most um i think that's the most prominent explanation for why we see this cancer difference that answers the question very interesting and staying with that topic of intergenerational imprinting that part was interesting to me as an insect biologist because parental imprinting is often used to be exploring differences in phenotype across nest mates within a colony is this a key uh domain of imprinting are imprinting effects widespread outside of this uh diametric model and what does this imprinting mechanism tell us about intergenerational epigenetic effects more broadly yeah so in terms of the effect of imprinting so i think that they're i i'm confident that imprinting plays a role in variation in this continuum but how much of a role i think is very much up in the air i don't think that we have a good idea of how many genes are actually imprinted uh the the number of genes that are said to be imprinted is about 200. both my advisor and i have expressed our doubts about this we think that it's probably more widespread than that and there are difficulties in picking up on something so you could have partial imprinting for example uh that it would be more difficult to pick up on uh but yeah so in terms of how this i i think it's pretty important uh christopher badcock who was one of the the authors on the on the first paper he's written a couple of books about this he's made the case that it's very important um i published a paper a couple years ago showing that people high in positive schizotype they tend to migrate more often or you know what i what we actually showed is that they have more intentions to migrate we've argued that that would lead to more migration but they at least say that they they are more likely to or they they want to move to other locations more now what you see in the animal literature is that dispersal is often a mechanism to reduce competition among among family members essentially you don't want to be competing with all of your siblings maybe that has something to do with it here it's a little different with humans because we're so cooperative but if we were being consistent with the animal literature that would be consistent with the animal literature on that on that kind of stuff yeah there's there's some very strange findings associated with this too i feel like are very controversial but i'll mention them anyways um so there was a book written so there was a paper and i i'm putting forward this stuff as very speculative so i want to be clear about that these are speculative things but there's a paper written a couple years ago

about imprinting making the case that paternally expressed genes in humans will be associated with more war-like behaviors so and this is because humans are thought to was the word be more patrilocal so it is the the women who tend to disperse more often which means that you're the the people who are in your group are more likely to be your male relatives they make an argument that that will lead to an increased uh group-like behavior in all this now a couple of findings that are associated with that which which are strange at least and that they might be associated with that there was a book published a couple years ago called the engineers of jihad and what they what they pointed out in that book is that people in engineering and technical professions are massively over represented among radical islamists and they're massively overrepresented among radical right-wingers which is a little surprising to me but the the data is very clear on that and i can't think of more war-like ideologies than those two you know uh if we're gonna be consistent with that with that theory now again i regard that as speculative but it's it's an interesting finding nonetheless we also see that people in the humanities uh who tend to be higher in positive schizotypes so they are the sort of artsy types they tend to be so people in the humanities are definitely more left-wing than other people and that's associated with ideologies that are sort of against war more often uh so so it's kind of an interesting finding um i would like to explore it more but i i don't know i would say it's definitely speculative anyways there's a lot of interesting implications of this that again have yet to be explored it's one of those experiments where you want a small sample size pretty rare for biology a few more areas i think that would be awesome to explore mixing it up a little bit how does what you're sharing and learning help us think about how to be respectful in our facilitation and communication in neurodiverse settings yeah so i think it's important to recognize that there are strengths and weaknesses associated with the with the sides of the continuum so you know i i think that we often we often tend to think that the the people who think very differently than us and it's easy to come you know it's easy to sort of tell if somebody uh has a very different thinking style than you i think that it's it's very easy to sort of think that that other thinking style is is less useful in whatever way um i think it's important to understand you know really the implication of this is that we're we're all good um and of course there you know there's an iq thing as well that plays into this but we're all we're all going to be relatively better or worse at picking up on different kinds of uh so people on that autistic side they're very good at picking up on these precise patterns right that are you know it's associated with engineering computer programming physics and mathematics and the more noisy patterns and the more not only noisy patterns but what's the word [Music] unstructured

problems right so unstructured problems uh that require these sort of ins so there is evidence that people high in positive schizotypy are better at solving insect problems for example like the nine dot problem if anybody is familiar with that um and so that's a that's an important difference and yeah i mean we're you know um i think that understanding this can help people to be a little more empathetic towards people who think differently than them yeah and also seeing like cognitive diversity amongst the nest mates in the context of the decision that the colony has to make can be harnessed and so that's very useful and definitely there's a long supply chain and last mile to respecting people in different specific contexts but definitely this helps link up a lot of the general themes like you mentioned explore exploit generalist specialist these are broader trends in biology and in statistics and machine learning and then the dance and the art and the science potentially is about bringing to bear these empirically informed and analytically derived in some situations frameworks and then meeting with the situatedness of someone in their own n equals one trajectory to really like make it work in that one situation yeah okay um there's a a sequence of questions i have on attention but first in the very beginning you mentioned um finding active inference and predictive processing framework kind of latching onto that and using that to continue building your research direction so what was it from a framework feature perspective that you felt was going to be providing a better explanation or prediction just how do you sense make amidst all of these decades and decades of bayesian brain predictive processing and coding active inference all these different threads in cognitive sciences and neuroscience what makes you go the way that you're going with your phd yeah i mean well you know i think i've read almost every theory of autism that there is you know like in terms of cognitive theories i mean obscure dissertations and all this stuff the reason i i came to this predictive processing understanding is just because it's the one that explains so of course i was coming at this from the perspective of the diametric model right and this is the only thing that i found that that adequately to my mind explains both sides of this continuum uh other theories of autism you know maybe they maybe they work for explaining some features of autism but i needed something that worked for both right and this was this was what worked for both and so when i uh when i realized that and i actually i sort of had that revelation while i was reading andy clark's book really uh surfing uncertainty um when i realized that i i realized that you know i i should learn more about this framework and so i sort of started consuming literature on predictive processing very uh obsessively at this point i think it's uh so another project i have you know i would like to work on is integrating some of the insights from evolutionary psychology with with predictive

processing i think that's an integration that could be really interesting but yeah that was basically it it's just that this uh that explanation the van der kroys and colleagues in their 2014 paper you know the explanation they put forward for autism not only explained autism but the opposite explained the other side for me and so that's kind of why i sort of latched on to this this framework and do you think following in that that there are unique explanations or unique experimental designs that would help us reduce our uncertainty about some of these contending theories yeah yeah i mean for the predictive processing uh theory you know we need more experiments on attention right because this is the most i think this is sort of the most straight well you know straightforward but it's it's the most obvious implication right if these are really differences in precision weighting we should be able to precisely uh look at differences in attention along this continuum and it's not always clear you know i i've read uh you know quite a bit of the experimental work on autism in relation to this there's been a lot of it it's not always clear what the predictions actually are right some sometimes um yeah sometimes the predictions are not straightforward you know this this certainly from a retroactive perspective i think it explains all this stuff really well uh in terms of predicting new stuff right it's it gets more complicated it just does you know um and so one of the things that we're going to do um for my phd is we're definitely going to look at the scope of attention it is the case that like people high in the in the trade of openness to experience uh they have a wider scope of attention right and we think this should be associated with positive schizotypy and there's good evidence that people with autism have a narrower scope of attention it's been called tunnel vision they've called it tunnel vision uh we want to demonstrate this in non-clinical populations and uh you know using the scale that we're making we want to demonstrate this and so you know that's one direction that we're going with this yeah i'll stop there i think that's all i have to say well awesome it leads to a few attention related questions that'll be great to dive into but but one reflection there is you described it as being able to retroactively connect a lot of dots through the usage of predictive frameworks however that is like one of those sides in the discussion today are we looking to um or which side are we looking to err on in that regression example was provided do we want the wavy line where the next batch of 10 points the wavy line is probably going to be totally off base because the initial wavy line was overfit or would we be okay with getting directional correctness but very little explanatory power like a high residual per point and the the optimum that science as this distributed system would be kind of navigating and weaving around that path of least action in that space would be the bayes optimal regression model which is actually what is described by

the bayesian information criterion or the aic its relative and those are statistical criteria that balance exactly that trade-off between a highly parameterized model that explains all the variants and then an under parameterized model too few parameters that explains maybe even a good chunk of the variation but then ultimately leaves a lot on the table when there's still like a a lot of um value to be gained by adding some more parameters so i was just seeing like that kind of bayesian information approach and then that's like the sweet spot but everybody moment to moment and year to year and life to life it's like a swarm weaving in that valley right and that is what constitutes part of our species level neurodiversity well you know i mean it's a good thing we have a lot of scientists working on this problem right so we can especially with autism you know so we can kind of go uh separate ways and meet in the middle a little bit you know we can kind of explore different different paths um yeah yeah go ahead let's talk a little bit about attention sure it's definitely a core term in active inference and it's um really one of the the tenants even though in reward-centric paradigms attention has been used like in neural networks that have attention and so it's interesting to see how even in a reward first paradigm still attentional features are brought in whereas active inference and predictive processing have a principled and analytically derived approach to attention so what is attention and what are the uh in a william james fashion what are the varieties of the attention experience yeah that's a tough question uh i mean i i've seen arguments you know within i think first in uh first and wrote a paper a couple years back arguing that there are these two uh different kinds of attention within within this framework right so there's like salience and then there is uh i forget what he called the other one but i'm not so sure that these are not really uh the same thing um you know what what it is to to pay attention to something my understanding within this framework is to treat you treat the input as if it's highly precise right you're treating it as if it's highly precise in a bayesian sense that means that you're treating it as if it's important it means it's having a large effect on updating your model and that includes you know because this is a hierarchical thing that also includes stuff that's generated by the the internal model right so you can give a high weight to you know a predict you know a prediction that's generated from your internal model so you're paying attention uh to what's going on inside of your own sort of internal model as uh as far as the varieties of attention yeah i mean i i don't think i have a good answer to that uh i don't i don't probably uh but that's but that's how i would define attention anyways uh with this within this model well it seems like a really interesting direction to see what kind of formalisms and models could underlie those um very gestalt capturing senses of attention like diffuse and laser and narrow but those

are like spotlight metaphors but where's the spotlight in the prediction and so it kind of crosses domains and so there's a lot of interesting um things there and yes what you said about uh attention being paid or salience in a bayesian statistical framework is exactly that like if there was a prior about the temperature and no attention was being paid to incoming thermometer readings then that prior would just play out whether it was a flat or whether it was expected to change from a top down hierarchical model whereas if a lot of attention were being paid to the thermometer readings in the extreme you just flip your best prediction to whatever the thermometer said less last and so then thinking back to those regressions that you showed they could be understood as in a static sense as being like what is the relationship between the person's height and how friendly they are to me and then that could be like sort of an all at once static regression about features of the world or like when it's raining what is the temperature like so just sort of across different instances regression that helps capture the direction and magnitude and reliability of some causal outcomes but it's also interesting to think about that x-axis as time and the y-axis is tracking again like temperature or some other thing in the world and there's like the high attention whiplash with the noisy measurements and it's like things are very volatile but they're being fit well but it's just feels like things are volatile and then the other extreme would be just that straight regression line where things are simple however there's a lot of variability and that actually also plays into certain bayesian work on for example like narrative and what is called conspiracy theory thinking simple that's the simple regression line that's narrative and then it's like well then why is it why is this why are these five examples in front of us not the case if that's totally the way things are in the world that a is associated with b but here's five a and it's not associated with b it's like right well there's error to explain there's residual but that doesn't dissuade me from believing that there is this relationship in the world that's true yeah it's interesting uh you know like the conspiratorial thinking is associated with positive schizotyping empirically which is what you would expect right you know it's a very simple explanation it explains a lot uh but it's you know it's a little bit unmoored from the actual data something like that it can tolerate high variance between the prediction and the observation and that can be accommodated in a framework where more attention is being paid to the narrative prior so the posterior is not updating that much seeing a hundred examples of something happening doesn't really update my likelihood on what i think is gonna happen next um yeah it was it's often the case that you know with like conspiracy conspiratorial thinking evidence against the conspiracy is taken as evidence for the conspiracy because it's just evidence that they're really really good at hiding it

right like uh you know like no evidence that this is going on is actually evidence of just how good they are at hiding it you know so you can incorporate almost any data into those kinds of models right it's an extremely uh yeah it's an extremely top-down biased way of thinking about the world and i think that's why it's associated with positive schizotypes it leads to one attention related question which is how do we train our attention or how do we shape our attention to be who we want to be well i i think i have two answers to that question uh the first one is meditation i think can help but also you know i i've had issues with with my own attention and one of the things that made me better at paying attention to things that i may not have wanted to pay attention to but that i really that i really need to is just having clear goals right it like gets your like if you have clear written goals it gets your dopaminergic system on on your side right so instead of uh you know so you get that little that little reward every time you do your physics homework or whatever which you may not want to pay attention to but if you're getting rewarded for it because you know you have a clear goal that you're moving towards it makes it easier to pay attention another thing i would say in in i think that's really it's under-appreciated how much our attention is related to our values you know when we value something it directs our attention towards relevant towards relation towards things that are relevant to the goal and so having clear goals is really important the other thing i would say is instead of you know instead of talking about changing uh or improving our attention is to put yourself into situations that are good for for your kind of attention style so you know i'm somebody who is not on the autistic side of that i would never have been an engineer okay or a physicist or anything like that you know it's not my thing um i had to put myself in a situation where i could use my intellectual ability the intellectual abilities that i have i love these big picture things i love seeing all the dots between all these disparate things you know that's what that's what i do i had to find a way to put myself in a position where i could use that um so some of it is is finding a way to get yourself in the right context that allows you to use your own attention style in the right way so it's not all about you know partly it's changing yourself but also partly it's changing uh the environment that you're in so that you can be who you are to some degree and that's definitely echoed in the ways that an active inference entity can reduce its variational or expected free energy which is it can update its generative model it can change its mind or it can engage in action it can change the world and i feel like that dialectic you know yesterday i tried to change the world today i tried to change myself and the way that we wouldn't want our actions to be malappropriated in the world or to be believing something falsely like that there isn't influence of our actions in the world but also it can be

extreme and unhelpful when people have far-ranging beliefs about their own role and so like social cognition and how that gets mediated with attention and shared attention thinking through other minds and joint attention general synchrony a lot of these active inference ideas are really in a stage where some of the continents in the map have been laid out and there's a lot of exploration to go fractal exploration to go to move from framing different things as like a deficit of reward to a perturbation of precision which also navigates the balance with pathology because we can say that bayes optimal cognition is occurring given the priors and the attention if this is the prior and here's how much of this type of attention that's being paid that's not a irrational or an even necessarily an incorrect it might be an outcome that isn't valued highly under some social criterion or another but then again that's that imperative to ask like what could be different in the person's developmental trajectory and their niche which includes their social niche so it is just opening up the discussion around these topics so it's a good time to be starting your phd in it yeah yeah definitely yeah with the with the psychopathology i mean i really love this you know the the uh this framework for thinking about psychopathology because there is that right like a lot of times what we label as psychopathology is actually you're you're working just the way you're supposed to um but i do think that there are there are times where we get caught in like these feedback loops right we get caught in these in these sort of uh mark miller has some good papers about this in relation to addiction and things like that you know we get caught in this uh you know what what they refer to as reciprocal narrowing it's like we're reducing prediction error in a local aspect but we're ignoring all of the all of the other areas in which we could be reducing prediction error so like if you're addicted to a drug of some kind if it's like a dopaminergic uh substance it's going to tell you that like you're doing great right you're reducing prediction error you know uh you know in an awesome way but uh everywhere around you you know it's just chaos right um so anyways so yeah i think this is a really cool framework for thinking about uh psychopathology in general and bring it into that extended if one cell were growing at the expense of others if it were having high fitness that's the kind of evolutionary lens if it were doing uh if it were modifying the signaling environment so it was getting exactly the hormone levels it expected even if those were potentially within a healthy range that might have deleterious consequences at the group fitness or free energy minimization or reward perspective and so we see kind of this braiding when things are working they can be analyzed with fitness reward precision and when things fail additionally we kind of see all of those different framings coalesce again and i think the interesting and informative space will be to

understand where those frameworks differ not just to justify or explore scientific hypotheses but truly to be in that gray zone and in that situation where it does matter how we act and whether we're justifying our actions with this or that flavor of evolution or reward or precision type thinking yeah it's it's uh yeah and it's it's exciting to sort of be at the beginning of this kind of stuff i mean i think we're really just getting started um in terms of you know understanding individual differences from a predictive processing perspective and psychopathology um you know like we're we're working on some stuff about psychosis too um and you know how positive schizotypal uh sort of the the sort of next step in this in this at least in the theoretical part of this one of the implications of the hypothesis i talked about in this paper so with autism and van der Kroy and colleagues talked about this in their 2014 paper prediction matching should take place at relatively low levels of the processing hierarchy right so like more the more concrete levels of the processing hierarchy uh for people with autism and that is sort of what we find it's a very detail-oriented style but with positive schizotypal you should see the opposite of prediction errors traveling farther up the hierarchy and what we think is happening in psychosis is basically like those prediction errors travel up to those at higher levels of the hierarchy like those are the those are the priors upon which many other priors depend and so if you disrupt those priors right you lose precision not only on those but on everything that they're dependent on and if that happens very suddenly you can get a massive increase in entropy and a massive increase in in behavioral and psychological entropy and kind of think that's what psychosis is um it's not entirely original to that idea but it just sort of talks about how it develops right because if we know that people who give a low weight to sensory input are are susceptible to it that tells us why they're susceptible right because those prediction errors are going to go up to those more high level priors if that makes sense is there anything else you want to add talk about ask i think i'm good man i really appreciate the opportunity to share my work and and have this conversation really good excellent best of luck with your continued research and education all right thank you daniel see you later bye